

Write a Python Program to Implement Decision Tree

Aim

To develop a Python program that implements the Decision Tree algorithm for classification.

Objective

- To understand Decision Tree classification.
- To classify data using supervised learning.
- To train and test a Decision Tree model.
- To predict the output class for new data.

Theory

A **Decision Tree** is a supervised machine learning algorithm used for classification and regression problems. It consists of a root node, decision nodes, and leaf nodes. The algorithm splits the dataset into smaller subsets based on the best attribute selected using measures like **Information Gain** or **Gini Index**. Decision Trees are simple to understand, easy to visualize, and widely used in Artificial Intelligence and Data Mining.

Algorithm

Step 1: Import the required libraries.

Step 2: Prepare the training dataset.

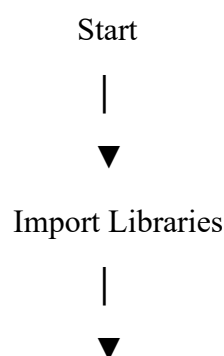
Step 3: Create the Decision Tree classifier.

Step 4: Train the model using the training data.

Step 5: Predict the class for the test data.

Step 6: Display the predicted output.

Flowchart



Prepare Dataset



Create Decision Tree Model



Train the Model



Predict Test Output



Display Prediction



Stop

Python Program

```
from sklearn.tree import DecisionTreeClassifier
```

```
X = [[0,0],
```

```
      [1,1],
```

```
      [1,0],
```

```
      [0,1]]
```

```
Y = [0,1,1,0]
```

```
model = DecisionTreeClassifier()
```

```
model.fit(X,Y)
```

```
prediction = model.predict([[1,1]])
```

```
print("Predicted Class =", prediction[0])
```

Sample Output

Predicted Class = 1

Result

The Python program successfully trained the Decision Tree model and predicted the output class.

Conclusion

Decision Trees provide a simple and efficient method for solving classification problems by learning decision rules from the training data.

```
C: > Users > Varshini M > OneDrive > Desktop > Here!!! > ARTIFICIAL INTELLIGENCE > VSC EXPs > EXP1.py > .
1  from sklearn.tree import DecisionTreeClassifier
2
3  # Training data
4  X = [
5      [0, 0],
6      [1, 1],
7      [1, 0],
8      [0, 1]
9  ]
10
11 # Target labels
12 Y = [0, 1, 1, 0]
13
14 # Create the Decision Tree model
15 model = DecisionTreeClassifier()
16
17 # Train the model
18 model.fit(X, Y)
19
20 # Test data
21 test_data = [[1, 1]]
22
23 # Predict the class
24 prediction = model.predict(test_data)
25
26 # Display result
27 print("Predicted Class =", prediction[0])
```